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ORIGINAL PAPER

Impact of prolonged rice cultivation on coupling relationship among C, Fe, and Fe-reducing bacteria over a 1000-year paddy soil chronosequence

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Abstract

Long-term soil chronosequences are valuable model systems for investigating pedogenesis and investigating the process of element coupling. Here, we assessed the coupling relationships among C, Fe, and Fe-reducing bacteria (*Anaeromyxobacter*, *Geobacter*, and *Shewanella*) in a paddy soil chronosequence of approximately 50 to 1000 years. Soils of the chronosequence originated from tidal marsh under nearly identical landscape and climate conditions. During 1000 years of rice cultivation, soil organic carbon (SOC) contents in surface horizons (0–20 cm) increased from 10.4 to 21.8 g kg^{-1}. In contrast, total Fe contents declined from 59.6 to 45.1 g kg^{-1} during the initial 50 years of paddy rice cultivation and then further decreased at a low rate of 0.004 g kg^{-1} soil $year^{-1}$ (equivalent to 10 kg ha^{-1} soil $year^{-1}$). Organically complexed Fe oxides (Fe_p) increased from 219 to 642 mg g^{-1} with increasing time of pedogenesis, but free total Fe oxides (Fe_d) and amorphous Fe oxides (Fe_o) declined at early stage of soil development, followed by a slow accumulation at later stages of the chronosequence. Gene copy numbers of *Anaeromyxobacter* and *Geobacter* increased from 4.6×10^5 and 3.6×10^6 copies g^{-1} to 3.8×10^7 and 3.6×10^7 copies g^{-1} dry soil with continuous paddy rice cultivation, while concurrently *Shewanella* gene abundance decreased gradually from 4.5×10^5 to 9.3×10^4 copies g^{-1} dry soil. Using structural equation modeling (SEM), different coupling relationships were observed among C, Fe, and Fe-reducing bacteria for the first 300 years of paddy chronosequence and thereafter. Overall, all Fe-reducing bacteria did not show consistent variation. With the stable microbial community and iron oxide fractions, the microbially mediated dissimilatory coupling relationship between C and Fe becomes simple during 1000 years of paddy soil development.

Keywords Paddy chronosequence · Fe oxides · Fe-reducing bacteria · C-Fe stoichiometry · Fe-reducing bacteria

Introduction

Paddy soils are hydromorphic soils formed under intense anthropogenic influence and generally subjected to fluctuating redox conditions induced by specific flooding and drainage regime (Kögel-Knabner et al. 2010; Zhang and Gong 2003). They store a substantial portion of the global agricultural soil carbon (C) pool and have a pivotal role in global C cycling

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(Pan et al. 2004; Liu et al. 2019a). Although paddy soils accumulate significant amount of organic matter (OM) in the topsoil (Liu et al. 2016a; Wang et al. 2015; Wissing et al. 2014), the processes leading to the retention of soil organic matter (SOM) under paddy management are not well understood, largely due to mechanistic uncertainty in the biogeochemical processes controlling the soil C.

Iron is one of the most abundant elements on the Earth's crust and plays an important role in various physiological and ecological soil processes (Boyd and Ellwood 2010; Kaiser and Guggenberger 2000, 2003). In surface horizons of paddy soil, frequent reduction-oxidation processes produce a diverse array of Fe oxides (Huang et al. 2018; Wen et al. 2018). Considering the high affinity for organic matter, Fe oxides strongly affect the C sorption reactions and the capacity of paddy soils to store C (Kögel-Knabner et al. 2010). The C-Fe association depends on the solid-phase characteristic of Fe oxides in paddy soils, e.g., size, structure, and surface area (Hassannezhad et al. 2008; Sodano et al. 2017), the Fe oxides' chemical reactivity (free, amorphous, or organically complexed ferric oxides), and their associated availability for microorganisms (Huang et al. 2018; Wen et al. 2018). In recent years, studies in element stoichiometry have gained increasing attention, as variation in C/nutrient ratios may aid in understanding the nutrient elemental balance in ecological interactions and processes, as well as providing an integrative nutrient framework linking to biogeochemical patterns on a global scale (Sterner and Elser 2002). Numerous studies have coupled the "iron redox wheel" with the biogeochemical cycle of C, N, S, and P individually in soils or sediments (Li et al. 2012). The Fe(II)-Fe(III) fluctuation is considered as an important redox process that links the reduction of Fe oxides to the oxidation (or mineralization) of OM; this implies a strong stoichiometric relationship between C and Fe in paddy soils. Unlike C/N/P element stoichiometry, however, studies that focus on the C/Fe stoichiometry in paddy soils are limited.

Pronounced spatially and temporally dynamics of the redox conditions make iron-transforming microbial communities vital in controlling the belowground C cycling processes in paddy soil (Dubinsky et al. 2010). When oxygen availability is limited, e.g., upon flooding and puddling, dissimilatory bacterial Fe(III)-reduction can transform Fe oxide minerals by rapidly reducing ferric iron and releasing ferrous iron into solution (Lovley et al. 2004). In some oxygen-scarce environments, the anoxic oxidation of C is almost entirely due to the reduction of Fe(III) by bacteria (Neubauer et al. 2005; Poggenburg et al. 2018). In addition, Fe(III) reducers can suppress methane (CH_4) production by competing with methanogenic archaea for acetate and hydrogen (Frenzel et al. 1999; Lovley and Phillips 1987; Teh et al. 2008). Hence, Fe-reducing microorganisms play a central role in SOM decomposition and greenhouse gas emissions from paddy soil.

Since Fe plays a central role in both biotic (microbial reduction or oxidation of Fe) and abiotic (e.g., proton-promoted, ligand-controlled Fe dissolution and mobilization, Fe adsorption and precipitation, and mineral transformation) soil processes (Burgin et al. 2011; Huang et al. 2018), a better understanding of microbially coupled C-Fe cycling processes in rice soil ecosystems is essential. Previous studies have investigated the impact of different fertilization regimes on the reactivity and stability of Fe oxides in paddy soils (Wen et al. 2018) as well as the effect of adsorbed or coprecipitated SOM on the microbial reduction of Fe-organic associations (Coward et al. 2018; Poggenburg et al. 2018). However, studies analyzing the microbially mediated dissimilatory coupling between C and Fe in paddy soils are rare. In particular, very little is known regarding the mechanism of shifts in the iron-reducing bacterial community induced by prolonged rice cultivation. Soil chronosequences provide a valuable tool for investigating the changes in different environmental properties and microbial thresholds under long-term rice cultivation (Liu et al. 2016b, c, 2019b). Previous studies have investigated the successive transformation and translocation of Fe oxides and Fe isotopic composition during rice cultivation on a millennial scale (Huang et al. 2018; Vogelsang et al. 2016). In the present study, a chronosequence approach was applied to investigate Fe-reducing microbial succession processes and the mediated coupling between SOC accumulation and iron oxido-reduction in paddy soil.

The objectives of the study were (1) to investigate the changes in Fe oxides and the Fe-reducing bacterial community, (2) to assess the stoichiometric relationships between C and Fe under prolonged rice cultivation, and (3) to estimate the potential effects of various Fe-reducing bacteria and iron oxide fractions on C accumulation during prolonged rice cultivation. We hypothesized that (1) after reclamation of initial tidal marsh, the quantities of various Fe-reducing bacteria and iron oxide fractions may significantly increase over the paddy chronosequence and (2) the coupling relationships of C, Fe, and Fe-reducing microbes change with paddy soil development. Therefore, the temporal dynamics of different forms of SOC, Fe oxides, and the Fe-reducing bacterial community were investigated along a paddy soil chronosequence consisting of six profiles derived from tidal marsh with a cultivation history ranging from 0 to 1000 years.

Materials and methods

Study site and soil sampling

The field sites were located in the Bay of Hangzhou near the city of Cixi, Zhejiang Province (30° 05′–30° 19′ N, 121° 09′–121° 31′ E) (Fig. 1) and were developed from comparable parent materials of paleo-deposits from the Yangtze River under comparable biogeographical conditions. Successive dyke building,

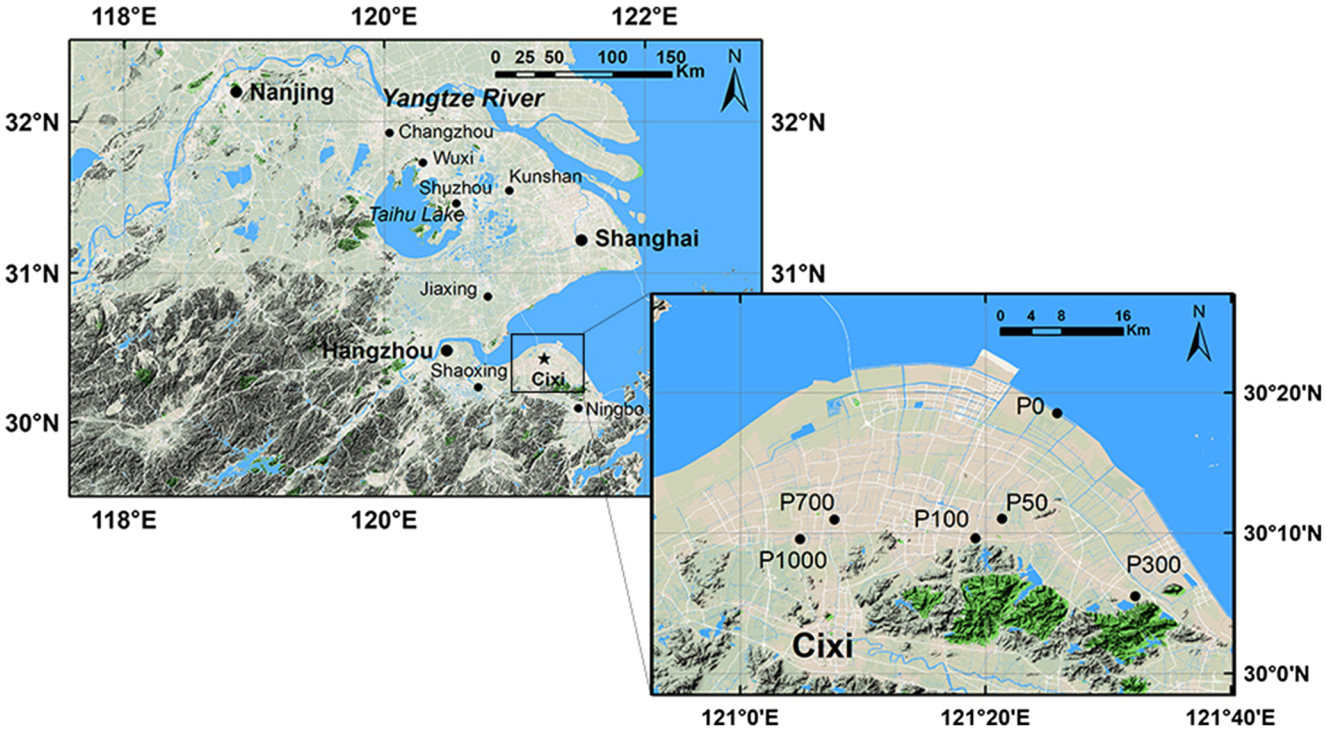


Fig. 1 Map of sampling sites. Locations are enlarged on the right of the figure, where the approximate spatial ranges of the soils of different ages are indicated. Soil age is represented by years since reclamation of wetlands

which started 2000 years ago, created a peninsula mainly consisting of Yangtze River sediment, on which the paddy soils of the chronosequence were formed. The study area has a typical north-subtropical monsoon climate, with a mean annual temperature of 16.3 °C and an annual precipitation average of 1325 mm (Cheng et al. 2009). The studied chronosequence was previously identified and pedologically characterized by Cheng et al. (2009) and investigated for morphological, mineralogical, and microbiological parameters by Kögel-Knabner et al. (2010).

Individual rice soils along the chronosequence with ascertained durations of rice cultivation were recognized based on dyke establishment history recorded in Cixi County Annals (with brief physical geographical information in Chinese available at www.cixi.gov.cn). An initial marsh soil before rice cultivation (P0) and paddy rice soils established by dyke construction approximately 50 years (P50), 100 years (P100), 300 years (P300), 700 years (P700), and 1000 years (P1000) ago (Fig. 1). All soils were located within 40 km of each other. Since they are situated in a relatively small area with a traditional summer rice-winter rape rotation, agricultural management has uniformly followed the central instructions of Technical Service Bureau since 1949. For all the paddy soils, the rates of fertilization and irrigation were similar among the fields (Cheng et al. 2009). All paddy fields are farther than 10 km away from the sea; thus, direct impact of sea water cannot occur. The surface soils are not influenced by groundwater due to the morphology of the paddy soils. Only in the first few years of reclamation might exist an influence of salt, but the surface soils are effectively desalinated after 50 years of freshwater irrigation and cropping (Kölbl et al. 2014).

Topsoil sampling

Site selection for the sampling of the individual rice soils of the chronosequence was based on a previous survey of soils in consultation with the Technical Service Bureau of the municipality of Cixi, Zhejiang Province. Soil sampling was performed before rice harvest in early November 2016, when the soil was still wet. For each soil in the chronosequence, five composite topsoil (0–20 cm in depth) samples, each one composed of 15 individual soil cores arranged in a Z shape, were collected from five randomly selected fields (each 500–700 m^2 in size) within the same dyke. For DNA extraction, small random aliquots of the samples were flash-frozen in a dry ice box and immediately stored at − 70 °C after shipping to the laboratory. One aliquot of the collected soil samples was sealed in a plastic bag, shipped to the laboratory within 2 days of sampling, and stored at 4 °C before soil analysis. The soil samples were cleared of gravel, roots, and visible plant detritus and then passed through a 2-mm mesh sieve for chemical analysis.

Soil physical and chemical properties

The soil pH and EC were determined at the soil-to-distilled water ratio of 1:5 (*w*/*v*) using a precision pH meter (Mettler Toledo Seveneasy, Greifensee, Switzerland) and a

conductivity probe (Thermo Scientific, MA, USA). The contents of SOC and soil total N (TN) were quantified using a CN analyzer (Vario EL III Elementar, Hanau, Germany, Rayment and Lyons 2011). Dissolved organic C (DOC) and N (DON) were measured using a TOC/TN analyzer (multi N/C 3100, Analytik Jena AG, Germany, Rayment and Lyons 2011). Soil NO_3^- and exchangeable NH_4^+ were extracted with 2 M KCl and determined by Auto Analyzer 3 (Bran+Luebbe, Norderstedt, Germany, Rayment and Lyons 2011).

To determine the total Fe content (Fe_t) in soils, samples were wet-digested with a concentrated acid mixture of HNO_3, $HClO_4$, and HF (Zhou et al. 2007). The free total Fe oxides (Fe_d) were extracted using a 0.1 M dithionite-citrate-bicarbonate (DCB) solution (Mehra and Jackson 1960), and the poorly crystalline Fe oxyhydroxides (Fe_o) were extracted with acid ammonium oxalate at pH 3, with a soil solution ratio of 1:50, and shaken for 4 h in the dark (García-Rodeja et al. 2004). The organically complexed minerals (Fe_p) were extracted with sodium-pyrophosphate (0.1 M) for 16 h (Carter and Gregorich 2007). The amount of Fe in the extracts was measured using inductively coupled plasma-optical emission spectrometry (ICP-OES; Vista MPX, Varian, Palo Alto, CA, USA). The concentrations of HCl-soluble Fe(II), Fe(III), and total Fe were measured after reduction of Fe(III) with hydroxylamine chlorhydrate as reported by Lovley and Phillips (1987). Briefly, about 0.5 g of sample was pipetted into 20 mL of 0.5 M HCl for 12-h extraction, and the extracted Fe(II) was determined using the *o*-phenanthroline colorimetric method (Tamura et al. 1974). Total extractable Fe was analyzed by the same procedure with the exception that the extractant was 5 mL of 0.25 M hydroxylamine hydrochloride. The amount of hydroxylamine-reducible Fe(III) was calculated as the difference between total extractable Fe and Fe(II) (Knorr and Blodau 2009).

The Fe free index is used to represent characteristics of the pedogenic process and was calculated as the proportion of Fe_d in Fe_t in the soil (Hu 2004), as follows:

$$\text{Fe free index} = Fe_d/Fe_t$$

The Fe activation index indicates the activation and transformation process of crystalline Fe oxides into amorphous Fe oxides (Blume and Schwertmann 1969), and it was calculated as follows:

$$\text{Fe activation index} = Fe_o/Fe_d$$

The Fe complexing index is used to represent the complexation degree of iron oxide and OM in the soil (Zou et al. 2008), and it was calculated as follows:

$$\text{Fe complexing index} = Fe_p/Fe_d$$

Soil microbial properties, DNA extraction, and real-time PCR assay

Microbial biomass C (MBC) and N (MBN) are indicators of the overall size of the soil microbial community, and they were quantified by the fumigation-incubation method and calculated using the factors of 0.45 and 0.54, respectively (Wu et al. 1990).

DNA from each of the 30 samples (6 sites × 5 replicates) was extracted from 0.45 g of soil with the PowerSoil™ DNA Isolation Kit (Mo Bio Laboratories, Carlsbad, CA, USA), according to manufacturer's protocol. The concentrations of the DNA extracts were determined at 260 nm using a spectrophotometer (Bio Photometer, Eppendorf, Germany), and integrity and size were checked using 1.0% agarose gel electrophoresis. The extracts were stored at − 70 °C until use.

Quantitative PCR was performed on a 7500 real-time PCR system (Applied Biosystems, Foster City, CA, USA) using SYBR Green as the fluorescent dye. The primers used to target the *Anaeromyxobacter*, *Geobacter*, and *Shewanella* 16S rRNA genes were Amyx-399F/Amyx-466R, Geo494F/Geo825R, and She-120F/She-220R (Himmelheber et al. 2008; Kato et al. 2010).

Fe-reducing bacterial community analysis

An aliquot (50 ng) of purified DNA from three samples taken from each soil was used as the template for amplification. The V4–V5 hypervariable regions of the bacterial 16S rRNA genes (*Escherichia coli* positions 515–907, Biddle et al. 2008) were amplified using Ex Taq HS DNA polymerase (TaKaRa Shuzo, Otsu, Japan) in triplicate. PCR conditions were as follows: 30 cycles of denaturation at 94 °C for 30 s, annealing at 55 °C for 30 s, and extension at 72 °C for 30 s, with a final extension at 72 °C for 10 min. The PCR products were resolved on an agarose gel, and the appropriate fragments were isolated and purified by the Agarose Gel DNA purification kit (TaKaRa Shuzo, Otsu, Japan). The purified PCR products were quantified using a spectrophotometer (Bio Photometer, Eppendorf, Germany). The PCR products from each sample were combined in an equimolar ratio in a single tube and run on a Roche FLX 454 pyrosequencing machine (Roche Diagnostics Corporation, Branford, CT, USA), producing reads from the reverse direction using R907.

The data were quality-filtered and chimera-checked as described previously (Fierer et al. 2008; He et al. 2010) using the Quantitative Insights Into Microbial Ecology (QIIME) workflow (http://qiime.sourceforge.net/, Caporaso et al. 2010). In brief, bacterial sequences with the same barcode were assigned to the same sample, and then barcode and primer sequences were removed. The bacterial phylotypes were identified using uclust (Edgar 2010) and assigned to operational taxonomic units (OTUs, 97% similarity). The

taxonomic identity of each phylotype was determined using the Ribosomal Database Project (RDP) Classifier (Wang et al. 2007). Across all samples, we obtained 168,315 quality sequences in total, including 5766–23,268 sequences per sample (mean = 11,221). We were able to classify 82.56% of those sequences at the phylum level (Table S1). Sequences with a length of over 200 bp were used by QIIME for this investigation, and the dominant length was approximately 400–420 bp. To correct for sampling effects, we used a randomly selected subset of 5500 sequences per sample for further analysis using QIIME. The 454-pyrosequencing data generated for this study have been submitted to the Sequence Read Archive (SRA) and are available under project SRP033387.

Statistical analysis

SPSS software version 20.0 for Windows (SPSS, Chicago, IL, USA) was used for standard statistical tests via one-way analysis of variance (ANOVA) on the soil biogeochemical and taxonomic data. Significant differences were determined by Tukey's HSD test at $P < 0.05$. Principal component analysis (PCA) of the Fe-reducing bacterial community based on the sequencing data was performed using SPSS 20.0 for Windows and was visualized using Origin Pro 9.1 (OriginLab, Northampton, MA, USA). Canonical correspondence analysis (CCA) was performed with CANOCO 5.0 for Windows (Microcomputer Power, Ithaca, NY, USA) based on gene abundance to identify the effects of soil physiochemical parameters on Fe-reducing bacterial communities in the paddy chronosequence. Structural equation modeling (SEM) was applied to analyze the direct and indirect relationships of biotic and abiotic variables among C, Fe, and Fe-reducing microbes using AMOS 21.0 (Amos Development, Spring House, PA, USA).

Results

Variation in soil physicochemical and microbial properties

The major physicochemical properties are summarized in Table 1. Within the 1000-year chronosequence, the soil pH decreased from 7.94 at P0 to 5.16 at P1000, and EC decreased from 3787 to 391 μS cm^{-1}, while SOC increased from 10.38 to 21.80 g kg^{-1} and TN increased from 1.17 to 2.34 g kg^{-1}. Concurrently, DOC and DON increased from 24.23 and 4.71 mg kg^{-1}, respectively, at P0 to values of 44.37–66.56 and 8.28–21.45 mg kg^{-1} in the various paddy soils (Table 1). MBC, MBN, and the concentrations of exchangeable NH_4^+-N and NO_3^--N in marsh soil (P0) were significantly ($P < 0.05$) lower than those in paddy soils (P50–P1000) but showed little and insignificant variation over the paddy soil chronosequence.

Variation in Fe concentrations

Total iron (Fe_t) ranged from 59.6 to 41.3 g kg^{-1}, which decreased with increasing duration of rice paddy cultivation. Ammonium oxalate-extractable Fe (Fe_o) and DCB-extractable Fe (Fe_d) decreased from 10.2 and 21.5 g kg^{-1} to 5.4 and 15.2 g kg^{-1}, respectively, during the initial 100 years of rice cultivation, while these parameters showed minimal changes thereafter. Pyrophosphate-extractable Fe (Fe_p) increased gradually from 220 to 643 mg kg^{-1} in the paddy soil chronosequence. Likewise, with increasing paddy soil age, HCl-extractable total Fe and Fe(III) increased from 3.7 and 2.4 g kg^{-1} to 6.1 and 5.2 g kg^{-1}, respectively, while HCl-extractable Fe(II) decreased from 1.3 to 0.9 g kg^{-1} (Table 2).

Significant changes occurred in the three chemostat indexes of Fe oxides during the initial 100 years of the paddy

Table 1 Physicochemical properties of the paddy chronosequence

Years of cultivation	pH	EC (μS cm^{-1})	SOC (g kg^{-1})	TN (g kg^{-1})	DOC (mg kg^{-1})	DON (mg kg^{-1})	MBC (mg kg^{-1})	MBN (mg kg^{-1})	Exchangeable NH_4^+-N (mg kg^{-1})	NO_3^--N (mg kg^{-1})
0	7.9 ± 0.2a	3787 ± 230a	10.4 ± 0.4c	1.2 ± 0.1d	24.2 ± 3.6c	4.7 ± 1.9d	205 ± 48d	8.2 ± 2.2c	0.9 ± 0.3c	15.5 ± 3.8b
50	7.7 ± 0.1a	564 ± 64b	15.6 ± 1.9b	1.7 ± 0.1c	44.4 ± 10.9b	8 .3 ± 2.0 cd	388 ± 57 cd	24.9 ± 5.2b	2.6 ± 0.3bc	46.9 ± 3.8a
100	6.8 ± 0.2b	408 ± 46b	19.5 ± 3.0a	1.9 ± 0.2c	49.9 ± 6.3ab	9.5 ± 2.6c	531 ± 92bc	26.7 ± 2.0b	3.7 ± 1.2ab	48.5 ± 9.7a
300	6.1 ± 0.4c	422 ± 43b	20.9 ± 1.7a	2.2 ± 0.2ab	64.6 ± 10.1a	15.3 ± 2.9b	553 ± 140abc	35.5 ± 9.5b	4.9 ± 1.2a	51.4 ± 2.8a
700	6.0 ± 0.1c	416 ± 37b	19.8 ± 0.9a	2.0 ± 0.1bc	59.2 ± 10.7ab	21.5 ± 3.8a	736 ± 143a	59.1 ± 11.3a	4.6 ± 1.5ab	59.0 ± 9.4a
1000	5.2 ± 0.3d	391 ± 42b	21.8 ± 1.9a	2.3 ± 0.1a	66.6 ± 13.5a	19.3 ± 2.9a	683 ± 100ab	55.4 ± 10.1a	4.6 ± 1.1ab	56.5 ± 6.4a

Significant differences between rice cultivation ages, indicated by different lowercase letters in the same column, were determined using one-way ANOVA followed by Duncan's multiple range test at $P < 0.05$. The data are shown as the means ± SD ($n = 5$)

EC electrical conductivity, *SOC* soil organic C, *TN* total soil N, *DOC* dissolved organic C, *DON* dissolved organic N, *MBC* microbial biomass C, *MBN* microbial biomass N

Table 2 The concentration of Fe oxide fractions in the soils of the paddy chronosequence

Years of cultivation	Fe_t (g kg^{-1})	Fe_o (g kg^{-1})	Fe_d (g kg^{-1})	Fe_p (mg kg^{-1})	HCl-Fe_t (g kg^{-1})	HCl-Fe(II) (g kg^{-1})	HCl-Fe(III) (g kg^{-1})
0	59.6 ± 0.6a	10.2 ± 1.4a	21.5 ± 1.5a	220 ± 25d	3.7 ± 0.2d	1.3 ± 0.1a	2.4 ± 0.1d
50	45.1 ± 0.8b	4.2 ± 0.2c	13.9 ± 0.5bc	241 ± 37d	4.5 ± 0.3 cd	1.2 ± 0.2ab	3.2 ± 0.3c
100	44.3 ± 0.6b	5.4 ± 0.3bc	15.2 ± 0.6b	326 ± 28c	4.9 ± 0.4bc	1.0 ± 0.1bc	3.9 ± 0.4bc
300	43.1 ± 1.8bc	5.6 ± 0.6b	13.8 ± 0.8bc	519 ± 23b	5.4 ± 0.3ab	1.0 ± 0.1c	4.4 ± 0.3b
700	41.6 ± 0.4c	5.8 ± 0.3b	13.1 ± 0.4c	552 ± 48b	5.4 ± 0.6ab	0.8 ± 0.2c	4.6 ± 0.6ab
1000	41.3 ± 1.2c	6.0 ± 0.3b	13.4 ± 0.3c	643 ± 50a	6.1 ± 0.4a	0.9 ± 0.1c	5.2 ± 0.5a

Significant differences between rice cultivation ages, indicated by different lowercase letters in the same column, were determined using one-way ANOVA followed by Duncan's multiple range test at $P < 0.05$. The data are shown as the means ± SD ($n = 5$)

Fe_t total Fe, *Fe_o* ammonium oxalate-extracted Fe, *Fe_d* DCB-extracted reactive Fe, *Fe_p* pyrophosphate-extracted Fe, *HCl-Fe_t* HCl-extracted total Fe, *HCl-Fe(II)* HCl-extracted ferrous iron, *HCl-Fe(III)* HCl-extracted ferric iron

chronosequence (Fig. 2). Thereafter, the Fe free index remained stable at approximately 0.32, whereas the Fe activation index logarithmically increased from 0.31 to 0.45. The Fe complexing index was low at P0 (0.04) and exhibited a logarithmic increase after 300 years of rice cultivation. The concentrations of Fe_p and HCl-Fe(III) were positively correlated with the SOC and DOC contents ($P < 0.01$, Fig. 3). The lowest C/Fe molar ratio was observed in the tidal marsh (P0), and it logarithmically increased with the paddy soil chronosequence (P50–P1000) (Fig. 4).

Variation in Fe-reducing bacterial community

In accordance with the pattern for SOC, soil *Anaeromyxobacter* and *Geobacter* gene abundance gradually increased with prolonged rice cultivation up to 1000 years. Gene copy numbers of *Anaeromyxobacter* and *Geobacter* were 4.6×10^5 and 3.6×10^6 copies g^{-1} dry soil at P0 but 3.8×10^7 and 3.6×10^7 copies g^{-1} at P1000, corresponding to 81- and 10-fold increase, respectively. In contrast, the *Shewanella* gene abundance in soil decreased gradually from 4.5×10^5 to 9.3×10^4 copies g^{-1} with rice cultivation (Fig. 5a). The frequency of *Geobacter* was the lowest (0.36‰) in the tidal marsh and increased with the duration of soil use (Fig. 5b). In contrast, the frequency of *Shewanella* significantly decreased from 12.4‰ at P0 to 0.06‰ at P50, while the frequency of *Anaeromyxobacter* significantly increased during the initial 300 years of rice cultivation, and then not significantly changed.

To examine changes in the overall Fe-reducing bacterial community along the chronosequence, PCA was performed based on OTU composition, including *Geobacter*, *Anaeromyxobacter*, *Shewanella*, Peptococcaceae, Bacillaceae, Clostridiaceae, *Bacteroidales*, and Desulfuromonadaceae (Table S2 and Fig. 6a). Three axes separated four groups (P0, P50–P100, P300–700, and P1000) in the plot, with the marsh soil being distantly separated from the paddy soils. Furthermore, we identified the relationship between the environmental properties and the Fe-reducing microbial

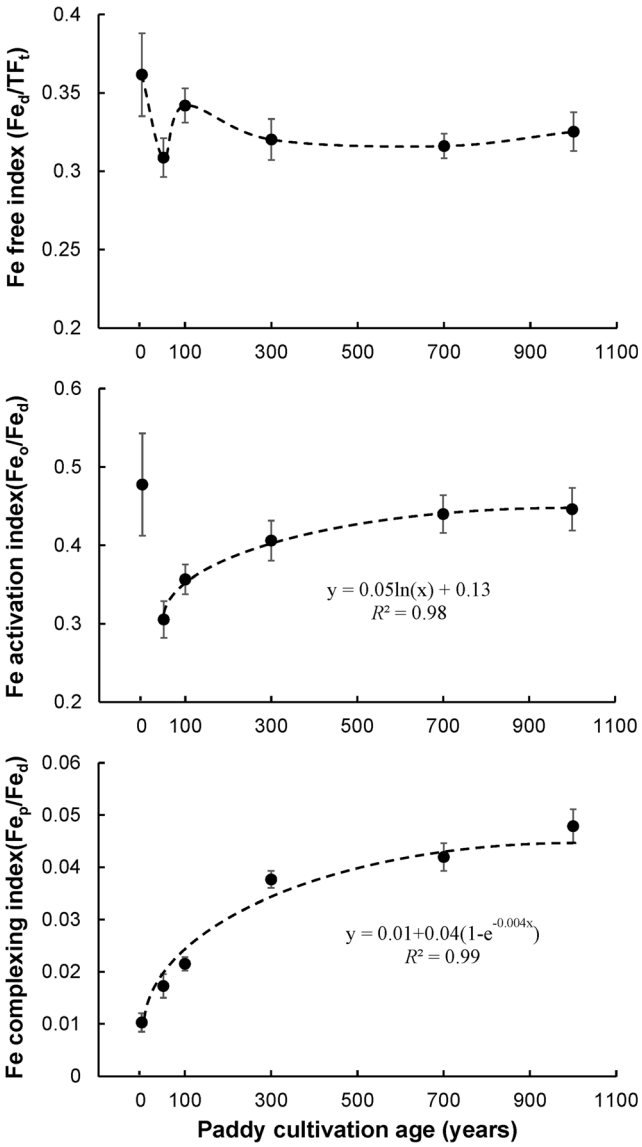


Fig. 2 Changes in chemostat indexes of Fe oxides with the duration of rice cultivation. Error bars indicate standard errors ($n = 5$)

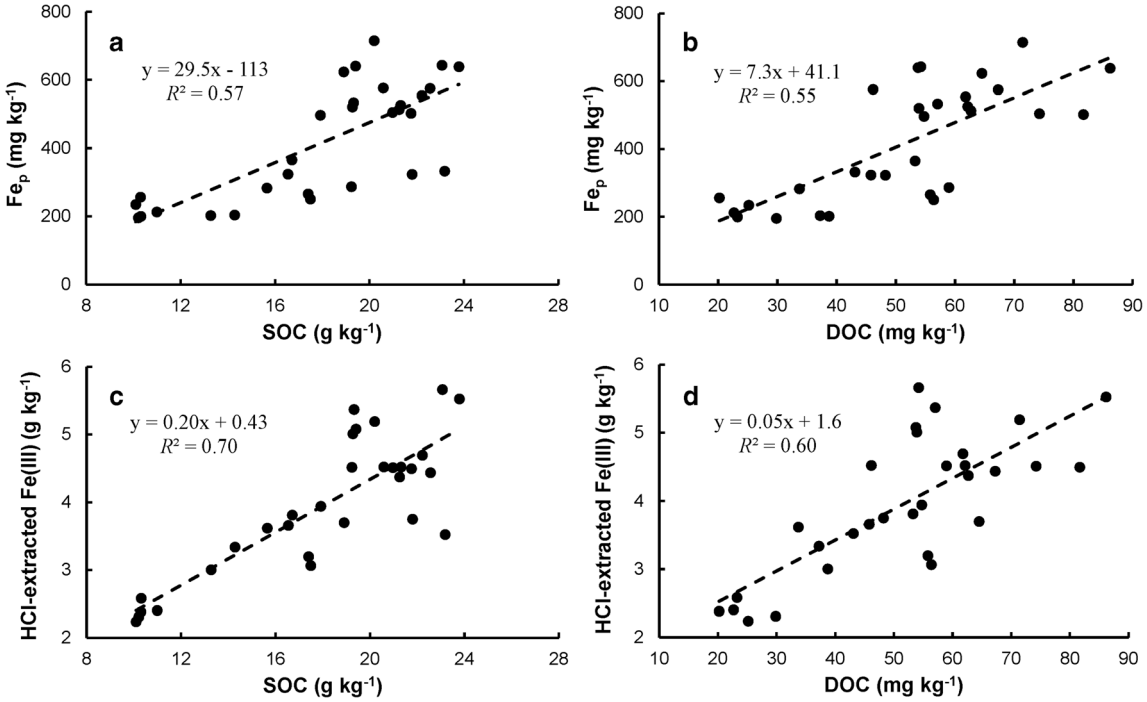


Fig. 3 Correlation of SOC and DOC contents with Fe_p and HCl-Fe(III) of the chronosequence soils. Error bars indicate standard errors ($n = 5$)

composition using CCA, which explained 81% of the total variance (Fig. 6b). CCA revealed that the SOC, TN, and DOC values were significantly correlated with *Anaeromyxobacter* and *Geobacter* abundances, whereas pH, EC, and HCl-Fe(II) were significantly associated with *Shewanella* abundance.

SEM was used to describe the direct and indirect pathways of variation in the different forms of soil C and Fe, as well as Fe-reducing bacteria abundance, during the two rice cultivation periods (P0–300: $\chi^2 = 113$, $P < 0.01$; P300–1000: $\chi^2 = 114$, $P < 0.01$; data from Tables 1 and 2 and Fig. 7). The model suggested obvious differences in the C-Fe-microbe coupling relationship between the two rice cultivation periods. For the first 300 years of rice cultivation, pH strongly (path coefficients > 0.7) influenced the Fe-reducing bacterial abundance. Thereafter, the interactions among C, Fe, and Fe-reducing microbes were less complex.

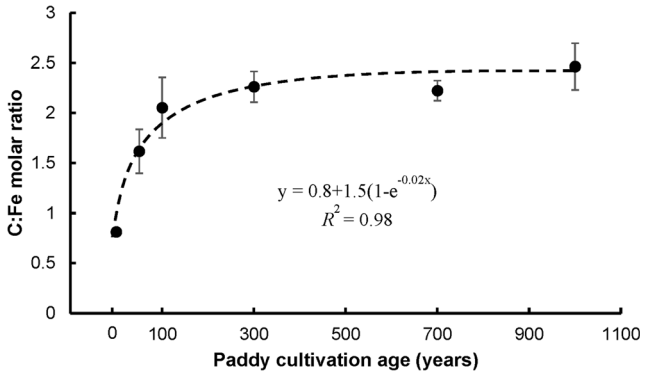


Fig. 4 Changes in SOC/Fe molar ratios with the duration of rice cultivation. Error bars indicate standard errors ($n = 5$)

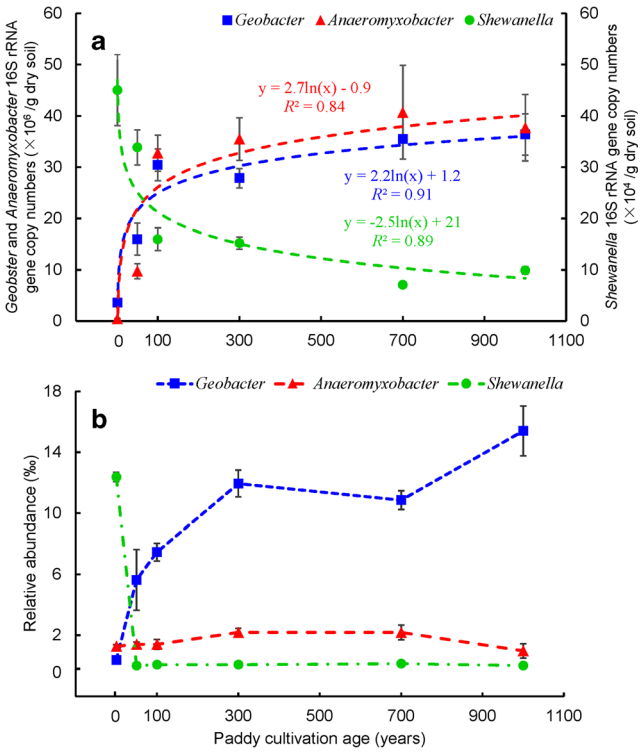


Fig. 5 Changes in 16S rRNA gene copy numbers (**a**) and relative abundances (**b**) of three Fe-reducing bacteria (*Geobacter*, *Anaeromyxobacter*, and *Shewanella*) with the duration of rice cultivation

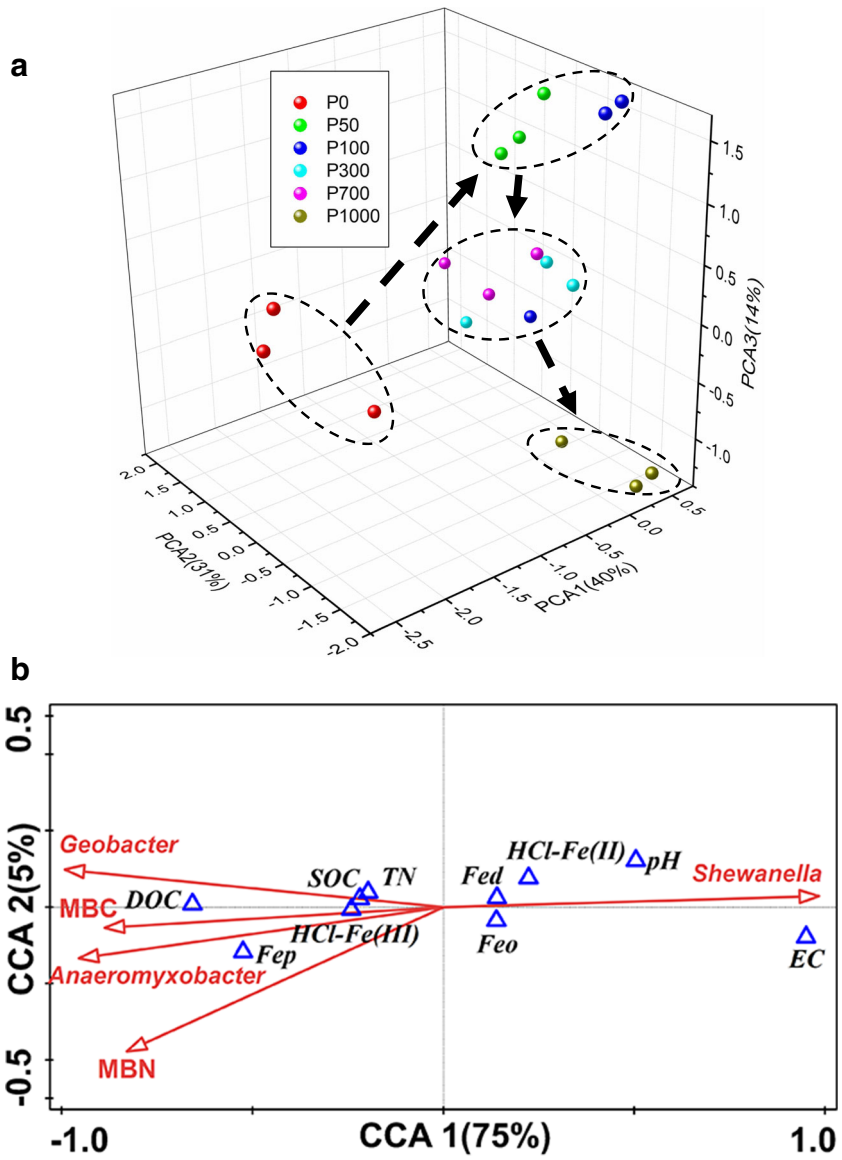


Fig. 6 Principal component analysis (PCA) based on OTUs of the Fe-reducing bacterial community in the chronosequence soils (**a**, data from Table S1). Canonical correspondence analysis (CCA) of the three Fe-reducing bacteria from the chronosequence soils (**b**, data from Tables 1 and 2 and Fig. 6). Gene copy numbers of *Geobacter*, *Anaeromyxobacter*, and *Shewanella* were used in the soil microbial community matrix (red arrows), and soil parameters [pH, SOC, TN, DOC, Fe_o, Fe_d, Fe_p, HCl-Fe(II), and HCl-Fe(III)] were used as constraining variables (blue triangles). Arrows indicate the direction and magnitude of measurable variables associated with microbial community structure

Discussion

Effect of cultivation time on soil Fe forms

In the Bay of Hangzhou, tideland reclamation and subsequent paddy cultivation have been practiced for thousands of years (Cheng et al. 2009). The conversion of tidal-terrestrial to a paddy system led to the gradual accumulation of OM from growing plants (Table 1). The rice cultivation and increasing inputs of chemical fertilizers generally lead to more dynamic pH and redox conditions (Guo et al. 2010; Kögel-Knabner et al. 2010; Lu et al. 2018), which affect the valence of iron, along with changes in the mobilization of Fe oxides, ultimately affecting the distribution of Fe oxide species.

The tidal alkaline environment impedes the loss of Fe from the topsoil, and thus, a high total Fe concentration at the initial stage of paddy soil evolution was observed by Huang et al. (2018). With the decreasing soil pH, total Fe concentration decreased at a low rate of 0.004 g kg^{-1} soil $year^{-1}$ (equivalent to 10 kg ha^{-1} soil $year^{-1}$) in the later stages of paddy soil evolution (Fig. S1). This is consistent with the observation that rice cultivation over a decadal time scale may enhance Fe depletion in acidic paddy soils (Han and Zhang 2013; Zhang and Gong 2003). The artificial continuous drainage during rice cultivation causes reducing conditions, thus resulting in the reductive dissolution of Fe oxides and leaching of Fe(II) to subsoils (Kirk 2004).

In this paddy chronosequence, all Fe oxide species changed rapidly in the first 50 years and then varied slowly over the later several decades and finally reached a plateau after 300 years of cultivation (Table 2). Fe_o declined in the first 50 years and then slowly accumulated with prolonged paddy

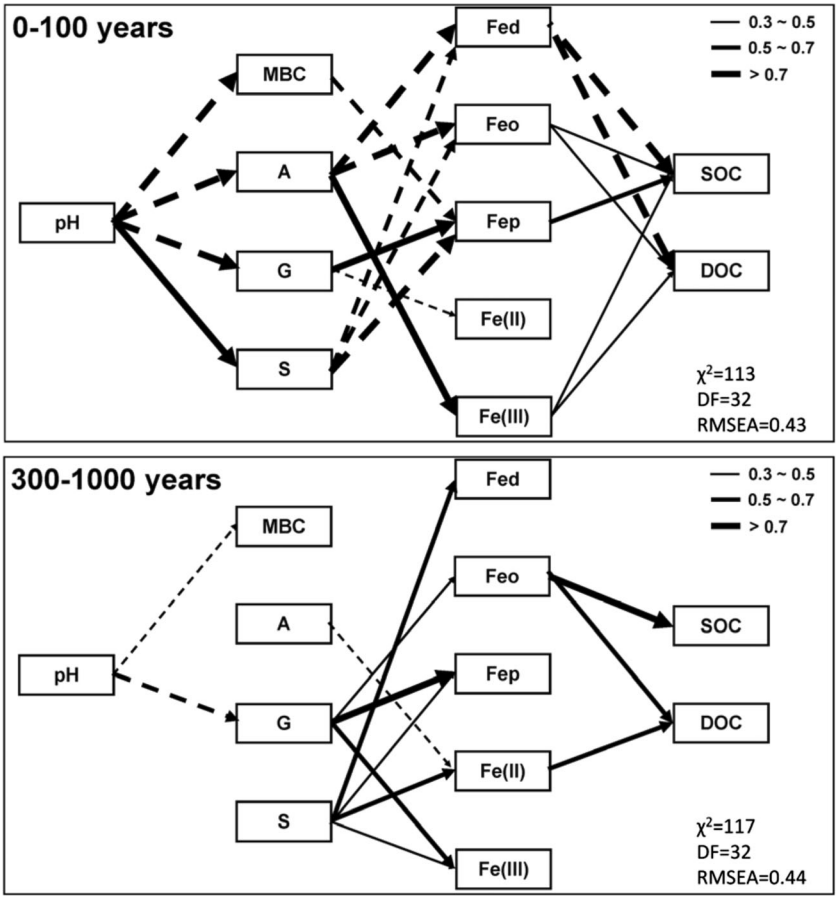


Fig. 7 Structural equation model (SEM) showing the hypothesized causal relationships among different fractions of organic C and Fe oxides, and the three Fe-reducing bacteria in the two stages (0–100 years and 300–1000 years). The width of the arrows indicates the strength of the standardized path coefficient. The solid black lines indicate positive path coefficients and dashed black lines indicate negative path coefficients

cultivation (Table 2). Such poorly crystalline Fe oxides exhibit highly reactive and sorptive capacities in aquatic and terrestrial systems, even at a low Fe concentration (Hiemstra and Van Riemsdijk 2009). Moreover, Fe_o is found to be highly susceptible to microbial reduction and dissolution in both soil and aquatic environments (Glasauer et al. 2003; Weiss et al. 2004), which may partly explain the observed trend in the Fe_o concentrations in this paddy chronosequence.

In the paddy chronosequence, Fe_d and the Fe_d/Fe_t ratio only changed during the first 300 years, reaching an almost steady state with little variations thereafter (Table 2 and Fig. 2). Both Fe_o and Fe_d contents decreased during the initial period of reclamation and rice cultivation, suggesting that these may not be stable Fe oxide forms in such environments. Concurrently, with a more drastic decrease in the Fe_o content, the Fe_o/Fe_d ratio (Fe activation index) sharply decreased in the initial 50 years and then reached a plateau after obtaining steady-state conditions (Fig. 2). It suggests that prolonged rice cultivation could result in relatively fewer low crystalline Fe oxides, which usually have a higher SOC sorption capacity and in turn preserve SOC (Zhou et al. 2009).

Pyrophosphate-extracted Fe (Fe_p) is mainly associated with OC in the soil and acts as an important C sink for paddy soils (Li et al. 2018). The concentration of Fe_p continuously increased with the duration of paddy management (Table 2). It has been well demonstrated that Fe_p, as a stable Fe pool, is hardly used by Fe-reducing bacteria in paddy soils, even though the abundance of some Fe-reducing bacteria increased with rice cultivation time (Fig. 5). Because of the different trends in Fe_d and Fe_p, the complexing index (Fe_p/Fe_d) increased in paddy soils over time (Fig. 2). These results demonstrated that prolonged submerged anaerobic conditions promoted greater steady-state percentages of Fe oxides, which is helpful in systematically understanding the mechanism of soil C stabilization.

Variation in C/Fe stoichiometry in response to prolonged rice cultivation

After the start of cultivation, the SOC and DOC contents linearly increased with an increase in Fe_p (Fig. 4). This strong correlation indicates the formation of Fe-OM complexes. Apart from the incomplete decomposition of OM under anaerobic conditions, Wissing et al. (2013) suggested that direct interactions with ubiquitous Fe oxides via chemical protection might contribute significantly to the long-term stabilization of OM. According to Sodano et al. (2017), the complexation of organic substances with Fe represents an important

stabilization mechanism for SOM in paddy soils. Likewise, SOC and DOC contents in the soil chronosequences were positively linearly correlated with HCl-extractable Fe(III) (Fig. 3). Besides the formation of Fe-associated OM by coprecipitation of Fe and DOM, contact of soluble Fe(II) with oxygen results in the formation of insoluble Fe(III), which concurrently sorbs ions and organic compounds (Fuller et al. 1993).

Recently, the C/Fe molar ratio has been considered to reflect the binding mechanism between SOC and Fe oxides in soil (Coward et al. 2018). When C/Fe > 1, SOC and Fe oxides may be associated via OM-accumulation mechanisms, such as coprecipitation, chelation, or aggregation (Wagai et al. 2013). In contrast, C/Fe ratios < 1 indicate the formation of Fe-OM associations via inner-sphere adsorptive mechanisms (Coward et al. 2018). In this chronosequence, before reclamation, the C/Fe molar ratio was < 1, whereas thereafter in all paddy soils, the ratios were > 1 (Fig. 4). Thus, we suggest that there were different mechanisms involved in the formation of Fe-OM associations in the original tidal soil and the paddy soils developed from it. Pham and Waite (2008) observed that Fe oxidation kinetics are faster at a low C/Fe ratio, which may explain larger contents of Fe_o and Fe_d in the initial tidal soil as compared to those in the paddy soils. The temporal trend of the DOC/Fe ratio was very similar to that of the C/Fe ratio (Fig. S2). Our results are consistent with prior observations that the retention of straw-derived DOM during coprecipitation with Fe strongly increased with increasing OC/Fe oxide ratios (Sodano et al. 2017). Moreover, Henneberry et al. (2012) indicated that DOC-Fe(III) oxide interaction produced a stable complex that did not release DOC back into the soil solution, regardless of changes in pH or Fe/DOC ratio. Coprecipitation may represent a key process in controlling DOC retention and is of particular interest for our understanding of OM stabilization and turnover mechanisms, implying the C source/sink functions of paddy soils.

Shift in Fe-reducing microbial communities in response to prolonged rice cultivation

During the initial period of reclamation and rice cultivation, C and N addition from plant growth and fertilizer was continuous. However, the microbial biomass was low (Table 1), and the consumption and utilization of C and N were limited, resulting in the accumulation of SOC and TN. Fifty years after the conversion of the tidal wetlands into paddy soils, MBC and MBN significantly increased, eventually reaching an equilibrium state due to the gradual stabilization of the soil environment (Table 1). Concurrently, the gene abundances of the three Fe-reducing bacteria (*Geobacter*, *Anaeromyxobacter*, and *Shewanella*) changed sharply within the first 50 years of cultivation (Fig. 4). During tidal-paddy evolution, the original mudflats were initially dehydrated to produce saline soil and finally reclaimed as paddy fields (Cheng et al. 2009). Changes in soil properties (e.g., salinity and pH) may influence the microbial biomass by causing osmotic stress (Egamberdieva et al. 2010). In addition, soil C and N are drivers of the microbial community during paddy soil cultivation (Atere et al. 2017; Ge et al. 2017; Liu et al. 2019c; Pietramellara et al. 2009).

Whether it is related to the electron transfer mechanisms of different types of Fe-reducing bacteria or not (Poggenburg et al. 2018), the abundances and compositions of *Geobacter*, *Anaeromyxobacter*, and *Shewanella* showed different responses to the impacts of prolonged paddy management (Fig. 5). *Geobacter* species have been reported to be the predominant dissimilatory Fe-reducing bacteria in anaerobic soils and sediments. Our results showed that the gene copy numbers of *Geobacter* and *Anaeromyxobacter* increased with increasing Fe(III) contents associated with the duration of paddy management (Fig. 5a). *Geobacter* can facilitate electron transfer to solid Fe minerals via direct contact or conducting nanowires (Malvankar et al. 2008) and are capable of oxidizing low-molecular-weight organic substances (e.g., acetate) as C sources by using Fe(III) as the terminal electron acceptor (Lovley et al. 1996). *Anaeromyxobacter* competes with *Geobacter* for iron oxides as electron acceptors. The relative abundance of *Anaeromyxobacter* remained low throughout the rice cultivation (Fig. 5b), showing the low competitiveness of *Anaeromyxobacter* with *Geobacter* (Hori et al. 2010). Gene copy numbers of *Shewanella* species decreased with rice cultivation age (Fig. 5a), and this change was positively correlated with soil pH and EC (Fig. 6b). *Shewanella* is able to excrete flavins as electron-shuttling molecules instead of relying on electron transfer via direct contact with the mineral (Kotloski and Gralnick 2013). Fe reduction by *Shewanella* is strongly related to the concentration of potentially usable electron-shuttling molecules of easily desorbable SOM (Poggenburg et al. 2018).

Despite significant changes in the gene copy numbers and the abundances of the three Fe-reducing bacteria in the initial 300 years, these changes gradually slowed with the soil cultivation time (Fig. 5). Many previous studies have suggested that longer paddy management results in a more stable and suitable environment for constant growth of certain bacterial populations (Liu et al. 2016b; Ding et al. 2017; Chen et al. 2018, 2019). In addition, the temporal pattern of Fe-reducing bacterial community succession was revealed by PCA (Fig. 6a). Over 85% of the variation in the community composition could be explained, indicating that the succession was primarily driven by long-term paddy soil development. This was consistent with the findings of Moore et al. (2010), who proposed that changes in soil chemistry over time led to shifts in subsurface microbial community across a marine terrace grassland chronosequence. We found that the succession of Fe-reducing bacterial communities is also influenced by soil

physicochemical properties (e.g., pH, salinity). Indeed, soil pH and salinity determine the spatial patterns of bacterial communities across global scales (Fierer and Jackson 2006; Lozupone and Knight 2007).

Effect of C-Fe-microbe coupling on soil C accumulation with rice cultivation

The accumulation and stabilization of OM in paddy soils depend on the low decomposition rate, mineral associations, and aggregate protection (Henneberry et al. 2012; Wang et al. 2015; Liu et al. 2016a). Under flooding, SOM decomposition proceeds at lower rates. Moreover, Fe oxides strongly influence the bioavailability of nutrients and C by adsorption (Eusterhues et al. 2014) or coprecipitation (Henneberry et al. 2012) due to the formation of Fe-organic associations. However, under anoxic environments, Fe-organic associations may be subject to microbial reduction (Lovley et al. 2004), which could expose formerly bound OM to microbial oxidation and decomposition (Canfield et al. 1993), suggesting a complex coupling relationship among C, Fe, and microorganisms.

Despite the interrelated nature of soil C and Fe biogeochemical cycling dynamics, the role of soil Fe oxides in C accumulation under prolonged rice cultivation has largely remained unexamined. The SEM, as a method for combining multiple regression, path, and factor analyses, has been widely used to distinguish interactions among multiple underlying and interacting factors in soil ecosystems (Fierer et al. 2009; Wade et al. 2018). In the current study, the SEM demonstrated that the Fe-reducing bacterial community and Fe oxide fractions play significant roles in C stabilization and accumulation dynamics within the paddy soil chronosequence (Fig. 7). Moreover, pH was coupled in the SEM approach to understand its direct and indirect effects on the Fe-reducing bacterial community in soils. In addition, across the entire data set in our study, soil C, Fe, and microbial species changed rapidly in the first 300 years of pedogenesis, and thereafter, plateaus of steady-state conditions were reached (Fig. 8). Thus, we divided the paddy chronosequence into two stages for analysis and investigated the different coupling relationship pathways of C-Fe-microbes.

The first stage (0–300 years) involves the transition from mudflat to paddy soil at the initial reclamation (Fig. 7). A small amount of tidal C accumulation in this period mainly came from aquatic algae and seaweed. After rice cultivation, high amounts of fertilizer were applied to promote rice productivity. The large residue return of rice contributed to the SOC accumulation. Bacterial communities gradually adapted to the changes in soil physicochemical properties, and microbial biomass increased quickly with the time (Liu et al. 2016b). Meanwhile, the rapid decline in pH strongly affected the abundances of the three Fe-reducing bacteria (Fig. 6b). Finally, Fe_o, Fe_p and Fe(III) had positive effects on the stabilization and accumulation of OC (Fig. 7). Under low DOC and Fe concentrations, poorly crystalline Fe oxides are reactive and a dominant adsorbent of nutrients in aquatic or terrestrial systems (Hiemstra and Van Riemsdijk 2009; Larsen and Postma 2001). Moreover, these poorly crystalline Fe oxides were found to be highly susceptible to microbial reduction in soil or aquatic environments (Glasauer et al. 2003; Weiss et al. 2004). This is consistent with our results that *Anaeromyxobacter* and *Geobacter* were negatively associated with Fe_o (Fig. 6b). In addition, iron-organic complexes can serve as nuclei for aggregate formation, reflecting the mechanisms of organic C stabilization via physical protection in developing soils (Silva et al.

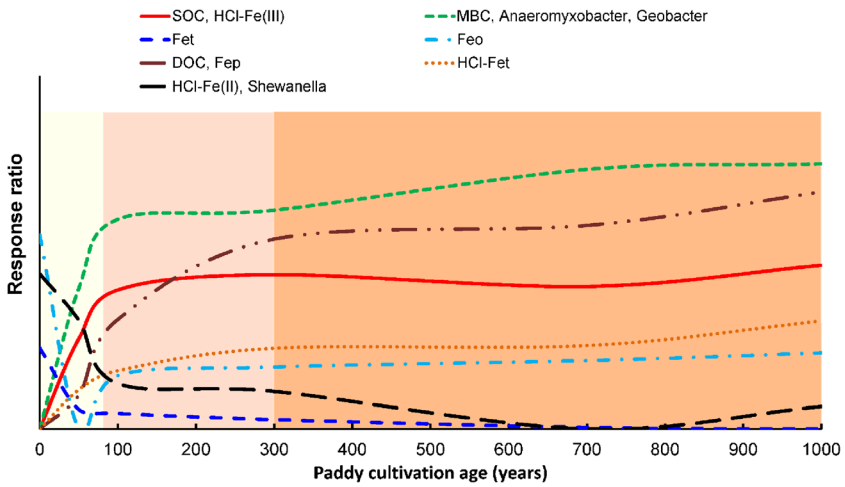


Fig. 8 Logarithmic response of soil Fe-reducing bacteria with regard to the redistribution of SOC, DOC, and Fe oxides along with rice cultivation duration. The response ratios were calculated as the ratio of a measured quantity in rice paddy soil under a certain duration of rice cultivation over that measured in the initial tidal marsh soil (Liu et al. 2016a). Red solid line, SOC and HCl-Fe(III); green dashed lined, MBC, gene copy numbers of *Geobacter* and *Anaeromyxobacter*; navy blue dashed line, Fe_t; light blue dashed line, Fe_o; brown dashed line, DOC and Fe_p; orange dashed line, HCl-Fe_t; black dashed line, HCl-Fe(II) and gene copy numbers of *Shewanella*

2015). The formation of soil aggregate here was consistent with findings in our previous research, which reported that the mean weight diameter of soil aggregates significantly increased within the initial 300 years of rice cultivation (Liu et al. 2016a).

The second stage began 300 years after the initiation of rice cultivation, and in this stage, the soil turned into a relatively stable paddy ecosystem, where the input and loss of C and N reached a relative balance and most physical, chemical, and biological properties were near a steady state (Fig. 8). The more stable pH only slightly affected the Fe-reducing bacteria (Fig. 7) owing to a reduction in selection pressure and adaption to the paddy environment (Liu et al. 2016b). Although the oxidation and reduction of Fe in paddy soil occurred throughout the succession, we found that various Fe pools were in the relative balance after 300 years of cultivation. In paddy soils, microbially mediated redox processes control the dynamics of various Fe pools, which are strongly related to the microbial species and accessibility of Fe. Thus, relatively stable microbial communities and soil environments would lead to more stable Fe redox processes than that occurred during the first 300 years after the reclamation of tidal marsh (Fig. 8).

In contrast to the first 300 years of rice cultivation, the SOC content was mainly affected by Fe_o and Fe(II) and less influenced by Fe_p (Fig. 7). Amorphous iron is regarded as a reactive Fe oxide due to its high surface area, which often accounts for most of the sorption and C stabilization (Wagai and Mayer 2007). Surface adsorption is well known, while coprecipitation has been recognized only recently as an important process responsible for C storage, particularly in hydromorphic soils (Henneberry et al. 2012). High concentrations of OM that often accumulate in anoxic paddy soils owing to the formation of organic-Fe(II) complexes (Jones et al. 2015) may coprecipitate with OM and lead to C retention under subsequent oxic conditions. Sodano et al. (2017) suggested that coprecipitation could result in a much higher C retention during the oxidation and hydrolysis of Fe(II) in the presence of rice-derived DOM. The higher C retention obtained by coprecipitation at high C/Fe ratios was probably due to the complexation of Fe(II) by low-molecular-weight organic acids, followed by oxidation and precipitation as C-rich metal salts (Sodano et al. 2017).

Conclusion

Within 1000 years of pedogenesis, gradual accumulation of SOC and TN was observed in paddy soil chronosequence. Prolonged frequent redox fluctuations led to decreased pH and total Fe content and the redistribution of Fe oxides. A longer history of rice cultivation tends to result in a strong stoichiometric relationship between SOC and total Fe contents in paddy soils. Moreover, the SOC and DOC contents in paddy soil chronosequence were positively linearly correlated with organically complexed Fe oxides (Fe_p) and Fe(III). The formation of Fe-OM associations plays a decisive role in OC stabilization and accumulation in paddy soils. The initial 300 years of paddy management resulted in rapid OC accumulation and was mainly affected by the dynamic concentrations of Fe_d, Fe_o, Fe_p, and Fe(III). Thereafter, the same management induced slow accumulation of OC which was associated with changes in Fe_o and Fe(II) contents. Overall, the coupling relationship of C-Fe-microbes in the first few years of rice cultivation was more complex than that observed after that of rice cultivation because the paddy ecosystem gradually stabilized over the period of rice cultivation.

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